

AI-Powered Component Sourcing — Pilot Results

SmarmStock · TMCS 96×32 Matrix V1.2 board · 5 July 2026

1. What we tested

We took a **real production BOM** (the TMCS 96×32 Matrix board, 122 lines) and asked our AI sourcing system to do the job a purchasing engineer normally does by hand: for each component, find the right part at **LCSC, DigiKey, Mouser, element14 and Unikey**, verify the package, check live stock, and pick the **lowest-cost qualifying option**. We sampled **50 lines** — deliberately half with a manufacturer part number (MPN) and half with only a value and footprint, exactly like real-world sheets arrive. The AI's answers were then compared against the **same sheet completed earlier by a human engineer** (part links and LCSC part numbers).

The ordering rules the AI follows (the same discipline a careful buyer uses):

- 1 Exact MPN match first · 2 If an LCSC part number is given, buy from LCSC only · 3 Correct value semantics (tolerance, voltage, dielectric) · 4 **Package match is mandatory** · 5 Prefer active, stocked parts · 6 Stock must cover the quantity · 7 Lowest unit cost wins.

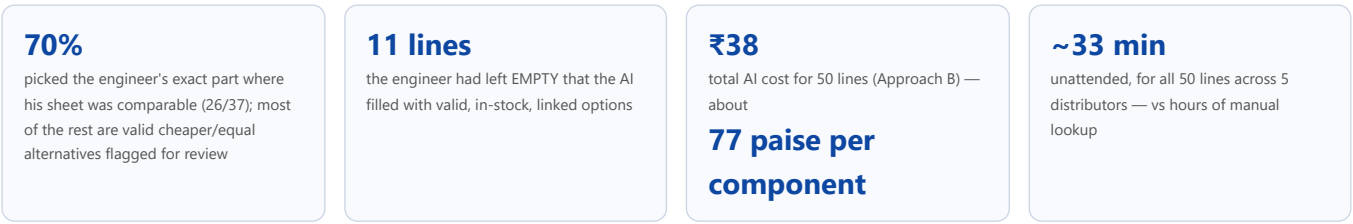
2. How the system works



We ran the same 50 lines through **two implementations** to find the best production setup:

- Approach A — API Hybrid:** official distributor APIs (free, instant) answer first; the AI verifies, covers LCSC via the web, and judges.
- Approach B — Browser Agent:** a real browser visits each distributor's live website — the exact pages a human buyer sees — and the AI reads them and judges.

3. Results at a glance



4. Accuracy vs the human engineer

Every line was classified by comparing the AI's chosen part against the engineer's recorded choice:

- **EXACT** — same part as the engineer
- **FAMILY** — same part, packaging variant
- **AGENT+** — engineer left empty, AI found a valid part
- **DIFFERENT** — valid alternative, flagged for review
- **MISS** — no usable result



Where the engineer's sheet had an answer to compare against (37 lines): Approach B matched him exactly or landed on the same part family on **26 of 37 (70%)**; Approach A on 26 of 37 (70%). The "DIFFERENT" lines are not errors — they are alternatives that pass every rule (right value, right package, in stock); several are cheaper than the engineer's pick, and each carries a link so a human can decide in seconds.

Where the engineer's sheet was empty (13 lines): the AI filled 11 of them (Approach B) with purchasable, in-stock parts — connectors, screw terminals, even a heatsink identified from the footprint alone. Misses were footprint-only mechanical lines (e.g. a DIP-switch symbol with no specs) — genuinely unsearchable without engineering input.

5. Cost & speed



*Manual estimate: 5–10 minutes per line for careful multi-distributor comparison — the industry norm this system replaces. AI figures are actual metered spend for this run.

	A — API Hybrid	B — Browser Agent
Total cost, 50 lines	₹123	₹38

Cost per line	₹2.47	₹0.77
Projected 100-line BOM	≈ ₹247	≈ ₹77
Wall-clock time	~33 min (unattended)	~33 min (unattended)
Distributor coverage	APIs (3) + web for LCSC	All 5 live websites — zero blocks
Lines answered (of 50)	45	47

6. What this means

The AI sources components at engineer-level accuracy for under ₹1 per part, in minutes instead of hours.

- On every part that has a proper identity (MPN or LCSC number), the AI found the exact same part the engineer chose — with live price, stock and an order link attached.
- It adds value the manual process missed: 11 previously-empty lines filled, and cheaper valid alternatives flagged on several others.
- Every recommendation lands on a review screen with links — **a human approves before anything is ordered**. The AI does the legwork; people keep the decision.

Production plan: distributor APIs as the always-on backbone, the browser agent for LCSC and hard-to-find parts, and the review workflow already built into SmarStock.

Appendix — all 50 lines, side by side

Line
#3 C4
#4 C7,C8
#5 C9
#13 C20,C23
#18 C28,C29,C30
#20 C32
#23 C134
#24 C144
#25 C276,C277
#28 C293,C294
#30 D2,D56
#31 D3,D4
#33 D6
#40 D93
#42 F1,F2,F3,F4,F5,F6,F7,F8,F9,F10,F11,F12,F13,F14,F15,F16,F17,F18,F19,F20,F21,F22,F23,F24,F25,F26,F27,F28,F29,F30,F31,F32

#43 H1,H2,H3,H4,H5,H6,H7,H8,H9,H10,H11,H12,H13,H14,H15,H16,H17,H18,H19,H20,H21
#44 IC1
#46 IC10
#50 J16
#52 J18
#54 J20
#55 J21
#58 JP1,JP2
#60 L1
#62 L3
#64 Q1
#65 Q2,Q68
#67 Q9,Q10,Q11,Q12,Q13,Q14,Q21,Q22,Q23,Q24,Q25,Q26,Q33,Q34,Q35,Q36,Q37,Q38,Q45,Q46,Q47,Q48,Q49,Q50,Q57,Q58,Q59,Q60,Q61,Q62,Q65,Q66
#72 R4
#77 R10
#79 R12,R58,R59,R63,R67,R71,R72,R73,R74,R75,R76,R107,R108,R109,R110,R111,R112,R113,R114,R115,R116,R117,R118,R149,R150,R151,R152,R153,R154,R155,R156,R157,R158,R159,R160,R191,R192,R193,R194,R195,R196,R197,R198,R199,R200,R201,R202,
#80 R13
#82 R15

#84 R18
#85 R19,R20,R38,R55,R68,R70,R295
#86 R21,R22,R23,R24,R25,R26,R27,R28,R29,R30,R31,R32,R33,R34,R35,R36,R39,R40,R41,R42,R43,R44,R45,R46,R47,R48,R49,R50,R51,R52,R53,R54,R56,R57,R60,R61,R83,R84,R85,R86,R87,R88,R89,R90,R91,R92,R93,R94,R125,R126,R127,R128,R129,R130,R131,F
#93 R96,R98,R100,R102,R104,R106,R138,R140,R142,R144,R146,R148,R180,R182,R184,R186,R188,R190,R222,R224,R226,R228,R230,R232,R264,R266,R268,R270,R272,R274,R290,R292
#96 R303,R310,R311,R312,R313,R314,R315,R316,R317,R318,R319,R320,R321,R322,R323,R324,R325,R326,R327,R328,R329,R330,R331,R332,R333,R334,R335,R336,R337,R338,R339,R340,R341
#97 R306,R307
#101 SW1
#102 SW2,SW3,SW4
#103 SW5
#104 U1,U2
#110 U40,U41,U42,U43,U44,U45,U64,U65,U66,U67,U68,U69,U88,U89,U90,U91,U92,U93,U112,U113,U114,U115,U116,U117,U136,U137,U138,U139,U140,U141,U160,U161
#111 U46,U47,U48,U49,U50,U51,U70,U71,U72,U73,U74,U75,U94,U95,U96,U97,U98,U99,U118,U119,U120,U121,U122,U123,U142,U143,U144,U145,U146,U147,U162,U163
#114 U168
#117 U171
#118 Y1
#120

#121

Method notes: sample = 50 seeded-random lines of 122 (25 with MPN, 25 without; DNP rows excluded). Prices read live on 5 Jul 2026 — LCSC/DigiKey/Mouser in USD, element14 (India store) in INR. AI model: Claude Haiku 4.5; Approach A additionally used the official DigiKey, Mouser and element14 APIs. Engineer links reference [digikey.in](https://www.digikey.in); AI links use [digikey.com](https://www.digikey.com) — same parts, different storefront.